Assignment 10: Data Scraping

Lex Clay

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1
library(tidyverse)
library(rvest)
library(lubridate)
library(here)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Set the year to 2024
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2

homepage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- Contact Person
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "Reginald Hicks", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3

#1. System Information Section

# scraping water system name
water_system_name <- homepage %>%
html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
html_text()
water_system_name
```

```
## [1] "Durham"
```

```r
#scraping contact person
contact_mensch <-homepage %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
contact_mensch
```

```
## [1] "Reginald Hicks"
```

```r
#scraping ownership
ownership <- homepage %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```r
#3. Water Supply Sources Section

#Maximum Day Use
maximum_day_use <- homepage %>%
  html_nodes("th~ td+ td") %>%
  html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

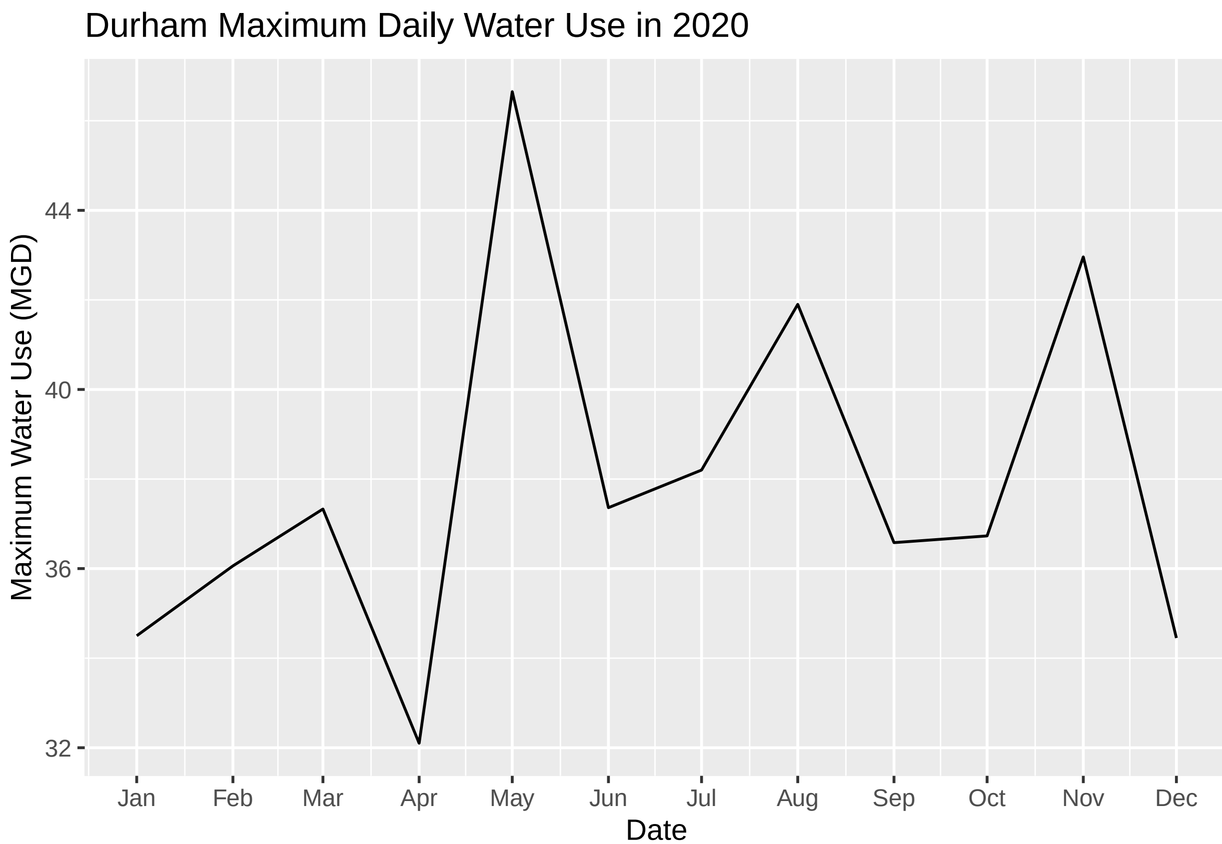
   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```r
#4

#frame with just numeric variablea
water.frame <- data.frame(
  "Month"=rep(1:12),
  "Year"=rep(2024,12),
  "maximum_daily_use"=as.numeric(maximum_day_use)
)

#adding string variables with mutate
water.frame.1 <- water.frame %>%
  mutate(
    Contact_Person = !!contact_mensch,
    Owner=!!ownership,
    Water_System =!!water_system_name,
    Date = my(paste0(Month, "-", Year))
  )

#5: Plot
water.plot <- ggplot(water.frame.1, aes(Date, maximum_daily_use))+
  geom_line()+
  labs(
    title =paste(water_system_name,"Maximum Daily Water Use in 2020"),
    y= "Maximum Water Use (MGD)",
    X= "Date"
  )+
  scale_x_date(
    date_breaks = "1 month",
    date_labels = "%b"
  )

print(water.plot)
```

```
## Ignoring unknown labels:
## * X : "Date"
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, "PWSID" and "year", that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```r
#6.


#automating before building function
stripped.report.url <- ("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=")
pwsid <- "03-32-010"
Year<- 2024

#function

fancy.scraper <- function(Year, pwsid){

  #website
```

```
 # fancy.report.url <- paste0(stripped.report.url, pwsid,"&year=",Year) #tried to run this first but it

  fancy.report.url <- read_html(paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=", pwsid

  #report variables
  water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  contact_mensch_tag <- "div+ table tr:nth-child(4) td:nth-child(2)"
  ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
  maximum_day_use_tag <-"th~ td+ td"

  #scrapified tags

  water_system_name <- fancy.report.url %>%
      html_nodes(water_system_name_tag) %>%
      html_text()
    water_system_name

  contact_mensch <-fancy.report.url %>%
    html_nodes(contact_mensch_tag) %>%
    html_text()
  contact_mensch

  ownership <- fancy.report.url %>%
    html_nodes(ownership_tag) %>%
    html_text()
  ownership

  maximum_day_use <- fancy.report.url %>%
    html_nodes(maximum_day_use_tag) %>%
    html_text()
    maximum_day_use


  max_water_master_frame <- data.frame(
    "Month"=rep(1:12),
    "Year"=rep(Year, 12),
    "Max Daily Water Use"=as.numeric(maximum_day_use)) %>%
      mutate(
        Contact_Person =!!contact_mensch,
        Ownership=!!ownership,
        Water_System_Name =!!water_system_name,
        Date=my(paste(Month, "-", Year))
      )

return(max_water_master_frame)

}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```r
#7

durham.water.2020 <- fancy.scraper(2020, "03-32-010")
glimpse(durham.water.2020)
```

```
## Rows: 12
## Columns: 7
## $ Month                <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
## $ Year                 <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 20~
## $ Max.Daily.Water.Use <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 4~
## $ Contact_Person      <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hick~
## $ Ownership           <chr> "Municipality", "Municipality", "Municipality", "M~
## $ Water_System_Name   <chr> "Durham", "Durham", "Durham", "Durham", "Durham", ~
## $ Date                <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 20~
```

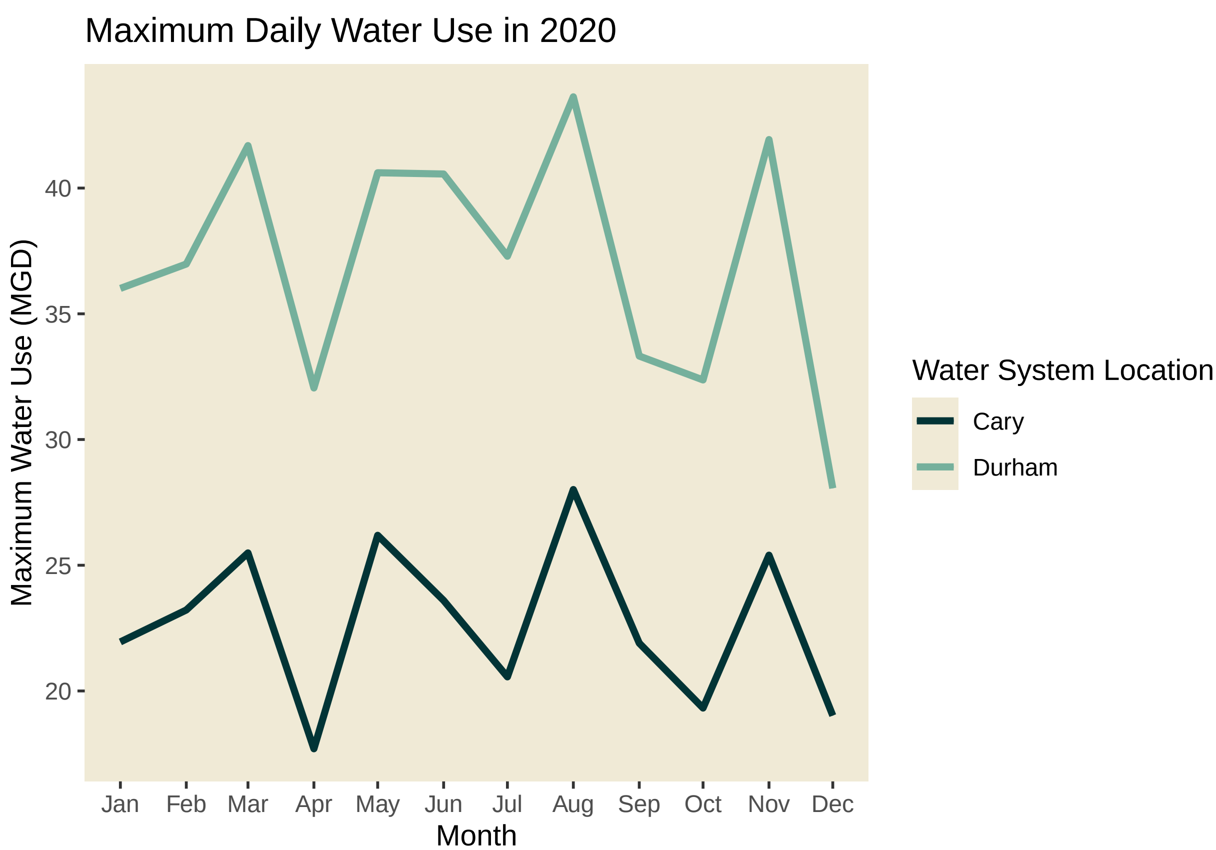
8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```r
#8

cary.water.2020 <-fancy.scraper(2020,"03-92-020")
view(cary.water.2020)

cary.durham.2020 <- bind_rows(durham.water.2020, cary.water.2020)

#plot it

cary.durham.2020.plot <- ggplot(cary.durham.2020, aes(Date, Max.Daily.Water.Use, color=Water_System_Name
  geom_line()+
  labs(
    title =paste("Maximum Daily Water Use in 2020"),
    y= "Maximum Water Use (MGD)",
    x= "Month",
    color="Water System Location"
  )+
  scale_x_date(
    date_breaks = "1 month",
    date_labels = "%b"
  )+
  theme(
    panel.background = element_rect(fill="#f0ead6"),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
  )+
  geom_line(size=1.25)+
  scale_color_manual(
    values = c("#023436", "#75B09C")
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
```

```
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(cary.durham.2020.plot)
```



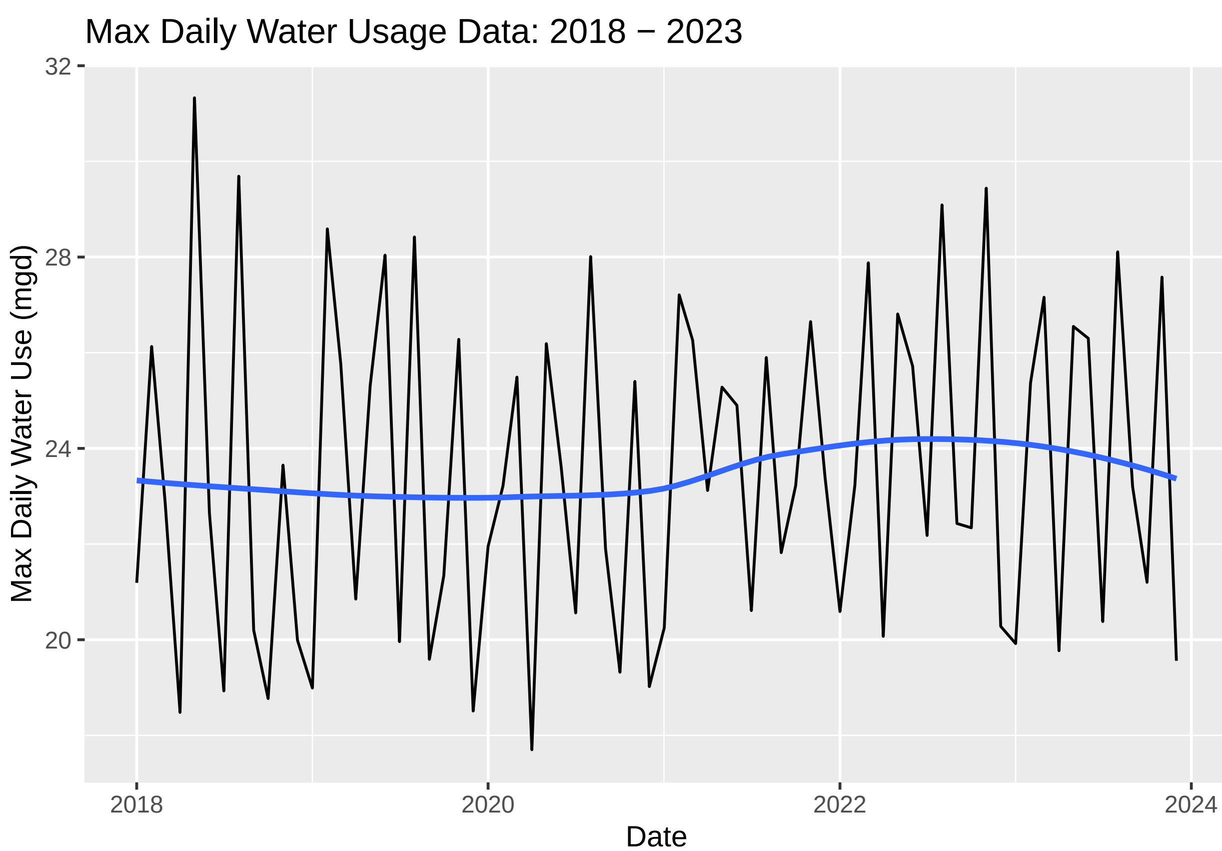
9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```
#9
The_Years <-rep(2018:2023)
Cary_PWSID_ID <- "03-92-020"


Cary_Almost_All <- map(The_Years, fancy.scraper, pwsid = Cary_PWSID_ID)

Cary_All <- bind_rows(Cary_Almost_All)

Cary_All_Map <- ggplot(Cary_All, aes(Date, Max.Daily.Water.Use)
```

```
                        )+
  geom_line()+
  geom_smooth(method = "loess", se=FALSE)+
  labs(
    title = paste("Max Daily Water Usage Data:", The_Years[1], "-",
                  The_Years[length(The_Years)]
     ),
    x="Date",
    y="Max Daily Water Use (mgd)"
    )


print(Cary_All_Map)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: > It is visually subtle, but the max water usage increases around 2021, peaks between 2022-2023, and then returns to pre-2021 levels.